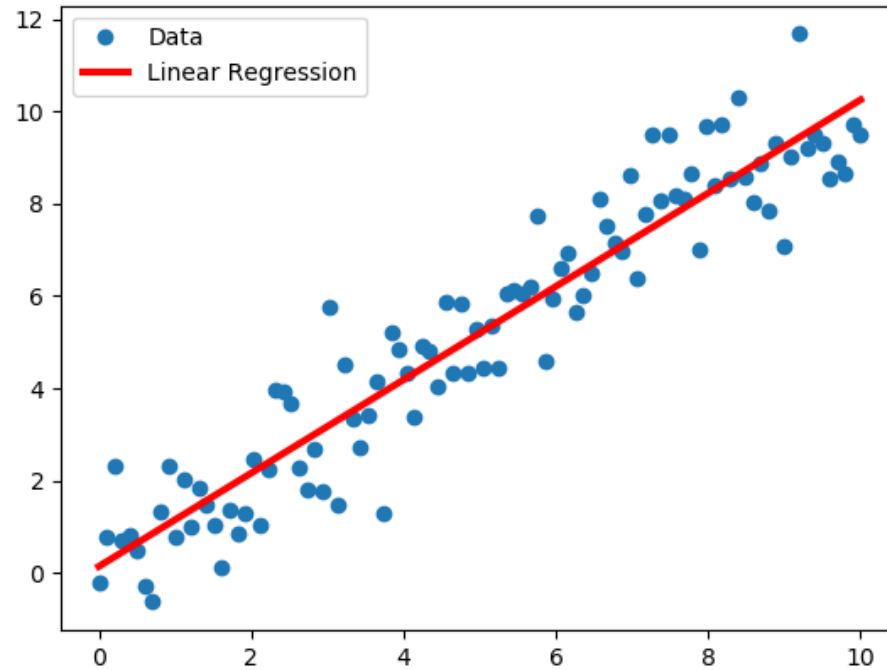




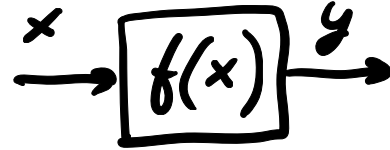
STATISTICAL LEARNING FOR AUTOMATION SYSTEMS

3. Linear regression

A.Y. 2022-2023

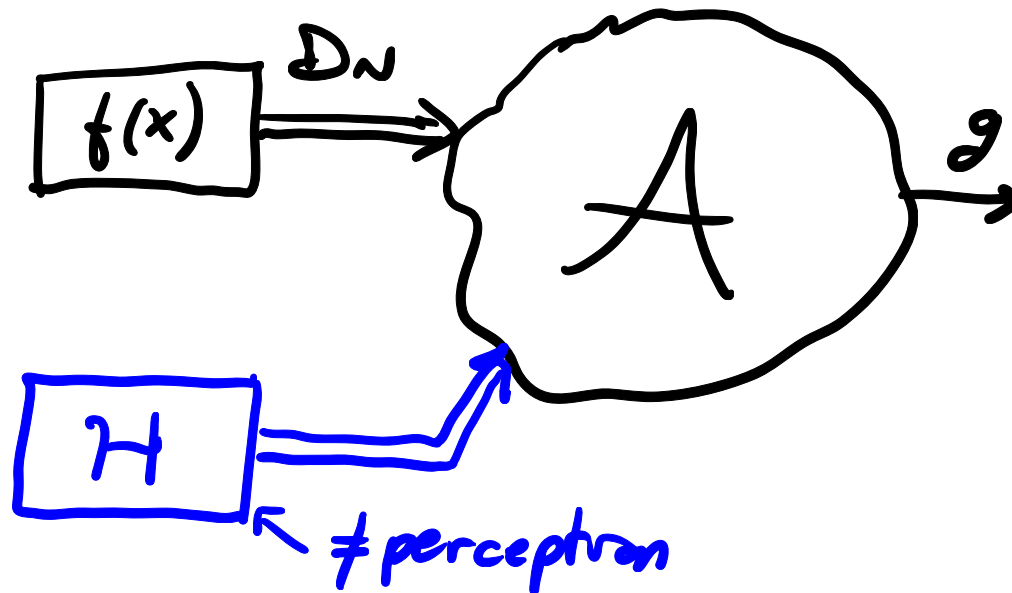


Prof. Simone Formentin



$$D_N = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}) \dots (x^{(N)}, y^{(N)})\}$$

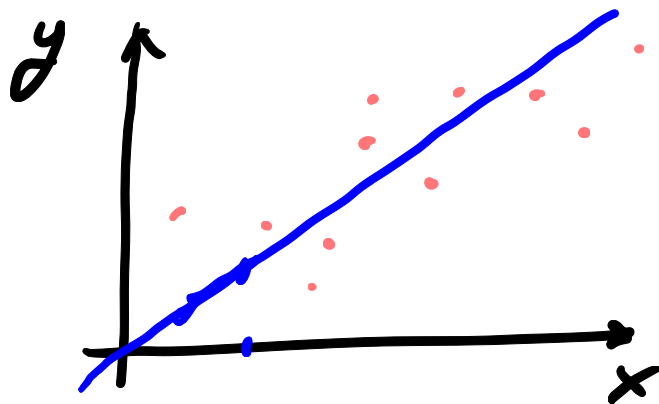
Unlike linear classification problems, here y is a number!



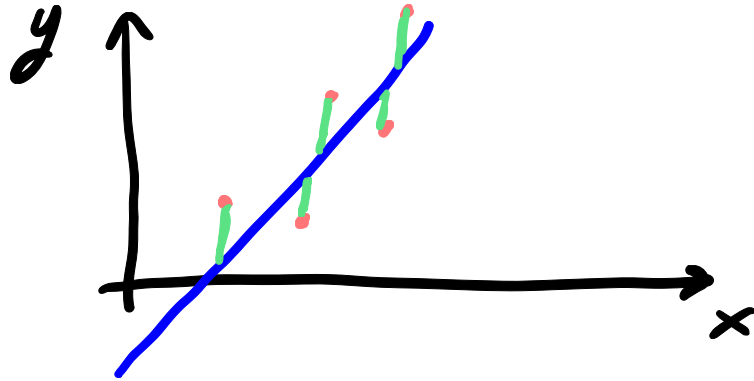
F. Galton studied the relationship between parents and children (1800)

Findings:

- ① A man's son tends to be as tall as his father
- ② The son's height tends to be close to the average height of the population



Key conclusion:
we should look at the population
to make the best prediction



Best model (line) $\equiv$ a line that minimizes a certain sum of the errors between the points and the line

$$\hat{g} = \arg \min_h E_{out}(h) = \mathbb{E} \left[(h(x) - f(x))^2 \right] \quad \leftarrow \text{unfeasible}$$

$$\min_h E_{in}(h) = \frac{1}{N} \sum_{i=1}^N (h(x^{(i)}) - y^{(i)})^2 \quad \leftarrow \text{feasible}$$

HYPOTHESIS SET: LINEAR MODELS WITH REAL-VALUED OUTPUT

1D-case: $y = w_1 x + w_0$

general case: $y = \sum_{i=1}^d w_i \cdot x_i + w_0 = \sum_{i=0}^d w_i x_i$

$$y = w^T \tilde{x}$$

$$\tilde{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_d \end{bmatrix}$$

← = 1

$$E_{in}(h) = \frac{1}{N} \sum_{i=1}^N (h(x^{(i)}) - y^{(i)})^2$$

$$E_{in}(w) = \frac{1}{N} \sum_{i=1}^N (w^T x^{(i)} - y^{(i)})^2$$

$\hat{w} = \underset{w}{\operatorname{argmin}} E_{in}(w)$
 parametric optimization problem

From now on, it is a PURE optimization problem

$$X = \begin{bmatrix} x_0^{(1)} & x_1^{(1)} & \dots & x_d^{(1)} \\ x_0^{(2)} & x_1^{(2)} & \dots & x_d^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ x_0^{(N)} & x_1^{(N)} & \dots & x_d^{(N)} \end{bmatrix}$$

$$X \in \mathbb{R}^{N \times (d+1)}, \quad Y \in \mathbb{R}^N$$

DATA MATRIX (REGRESSOR), (SCALAR OUTPUT)

$$Y = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(N)} \end{bmatrix}$$

$$w = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix}$$

$$E_{in}(w) = \frac{1}{N} \sum_{i=1}^N (w^T x^{(i)} - y^{(i)})^2$$

$$E_{in}(w) = \frac{1}{N} \| Y - Xw \|^2 = \frac{1}{N} (Y^T Y + w^T X^T X w - 2w^T X^T Y)$$

- necessary condition for optimality:

$$\nabla E_{in}(\hat{w}) = 0 \quad (\text{gradient is null})$$

- additional condition for $\hat{w}$ to be a minimum point:

$$\left. \frac{\partial^2 E_{in}(w)}{\partial w^2} \right|_{w=\hat{w}} > 0 \quad \text{positive definite}$$

Def. A matrix $M_{(d+1) \times (d+1)}$ is positive definite ($M \succ 0$) if $x^T M x > 0, \forall x \neq 0$

LINEAR ALGEBRA TRICKS:

$$\textcircled{1} \quad \nabla_w (w^T A w) = (A + A^T) w, \quad A \in \mathbb{R}^{(d+1) \times (d+1)}$$

$$\textcircled{2} \quad \nabla_w (w^T b) = b, \quad b \in \mathbb{R}^{(d+1)}$$

COMPUTATION OF THE GRADIENT OF $E_{in}(w)$

$$E_{in}(w) = \frac{1}{N} \left(w^T X^T X w + Y^T Y - 2 w^T X^T Y \right)$$

$$\nabla E_{in}(\hat{w}) = \frac{1}{N} \left(2 X^T X \hat{w} + 0 - 2 X^T Y \right) = 0$$

$$\frac{2}{N} \left(X^T X \hat{w} - X^T Y \right) = 0$$

necessary
condition for
optimality of $\hat{w}$

$$(X^T X) \hat{w} = X^T Y$$

normal
equations

COMPUTATION OF THE HESSIAN OF $E_{in}(w)$

$$\nabla E_{in}(w) = \frac{2}{N} (X^T X w - X^T Y)$$

$$\frac{\partial^2 E_{in}(w)}{\partial w^2} = \frac{2}{N} X^T X$$

second condition
for $\hat{w}$ to be a
minimum point

$$X^T X \succ 0$$

If $X^T X \succ 0$, then we can compute $(X^T X)^{-1}$ and the normal equations imply

$$\hat{w} = (X^T X)^{-1} X^T Y$$

ORDINARY
LEAST
SQUARES

$$\underbrace{(X^T X)^{-1} X^T}_{\text{pseudo-inverse of } X} = X^+$$

Analytical expression of the optimal model parameters that depend on data only!

REMARK: $X^T X \succ 0$ by construction ($X^T X = 0$ only if the data are not informative enough)

What about generalization?

Theorem:

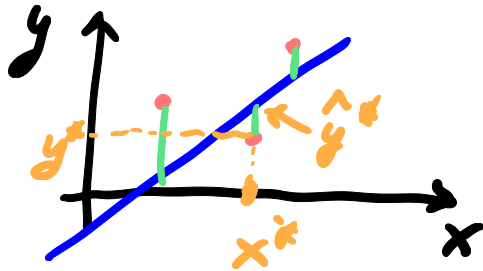
$$E_{out}(w) = \mathbb{E}[(w^T x - f(x))^2]$$

$$= E_{in}(w) + O\left(\frac{d}{N}\right)$$

dimension of x

size of the dataset

REMARK



$$\hat{Y} = X \hat{w} = X \underbrace{(X^T X)^{-1} X^T}_{H} Y = H \cdot Y$$

H : "hat" matrix